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This is a walkthrough lab, there will not be a corresponding video.

Activity 1: How to read an offline Registry file with Windows Registry Recovery.

Read) ARTIFACTS IN THE REGISTRY Aside from containing configuration settings for a Windows-based system, the Windows Registry contains a wealth of data about system usage. Users might think twice if they knew how much information is retained in the collective set of files known as the Registry. Since manipulating the Registry is something the typical computer user does not do, the data found in the Registry is considered inherently more reliable (although not perfect) than user data files. Two of the 4N6 goals with analyzing the Registry are: **1. Knowing what data stored in the Registry 2. Retrieving the data in a usable format**. On Windows-computer systems with large storage capacities, some investigators find examining the Registry to be an effective triage, because it is easier to recover all of the Registry files and focus on them rather than physically acquiring a multi-terabyte drive.
The Windows Registry is compromised of the following data files:
C:\Windows\system32\config\default C:\Windows\system32\config\SAM
C:\Windows\system32\config\SECURITY C:\Windows\system32\config\software
C:\Windows\system32\config\system C:\Users\username\NTUSER.DAT (for each user profile on the system)
When the files are loaded into memory, the Registry takes the form of:
HKEY_CLASSES_ROOT
HKEY_CURRENT_USER
HKEY_LOCAL_MACHINE
HKEY_USERS
HKEY_CURRENT_CONFIG
The activities presented here will examine a number of popular Registry entries, but clearly not all artifacts.

Instructions Activity 5 (DO)

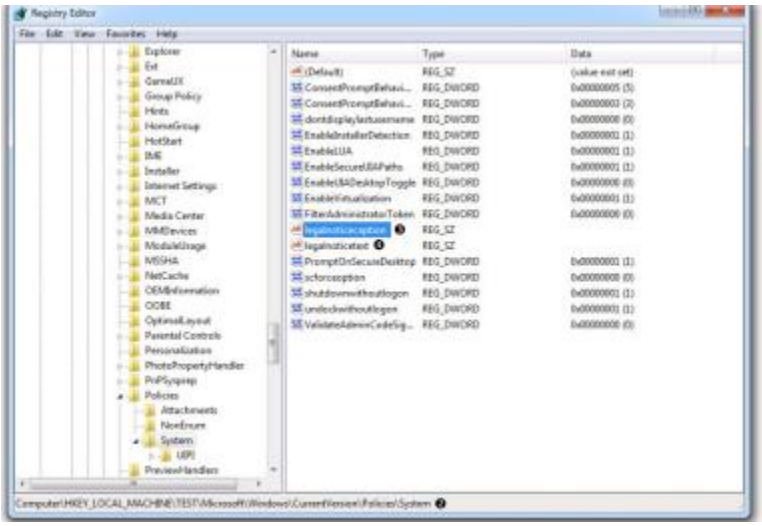
Should be installed already, but if not ……
Reading Offline Registry Files with Regedit
Product: Regedit Manufacturer: Microsoft Corporation Web site:
https://msdn.microsoft.com/en-us/library/windows/desktop/ms724871(v=vs.85).aspx

Warning: Please be extremely careful, when using Regedit. Changes made to the active Registry can cause unstable conditions in Windows.
1. Download the file called "RegistryFiles-1.zip" from filePack and extract the contents of the compressed file to your desktop.
2. Open a command prompt on a Windows computer.
3. fire up RegEdit
4. When the Registry Editor launches, ensure all keys are collapsed.

5. In the Regedit window, left click on HKEY_LOCAL_MACHINE. It will highlight. Do not open it.
6. From the main menu select “File” and then select “Load Hive…” from the pull-down menu. (If HKEY_LOCAL_MACHINE is not highlighted, this menu item will not appear.)
7. Browse to the directory on the desktop with the Registry files -1 retrieved from filepack. Select the file called SOFTWARE. When loading the file, you will be prompted to enter a name in the “Load Hive” window. Enter the name “TEST” and click the “OK” button.

8. Expand HKEY_LOCAL_MACHINE.
The loaded hive will appear with the name TEST. Confirm the logon banner contained within the Windows Registry of the TEST hive by navigating down to the following Registry key:
HKEY_LOCAL_MACHINE\TEST\Microsoft\Windows\CurrentVersion\Policies\System

9. After navigating down to the key, the path will be displayed in the lower left corner of the screen as shown below:



Notice two keys: legalnoticecaption and legalnoticetext. The former would contain the text value, which appears in the title bar of the consent banner. The latter is the actual message contained within the body of the consent banner.

10) What consent banner is shown on this computer? *No consent banner is displayed based on the empty values in this Registry file – you know this by double clicking the legalnoticecaption and finding “value data” empty. (In this hive, the consent banner has been removed and nothing will be displayed at logon. The absence of the banner may cause legal concerns during the examination of corporate assets. In this example the absence of data is a finding.) If a banner is found, it proves the user was informed of policies that were listed.*

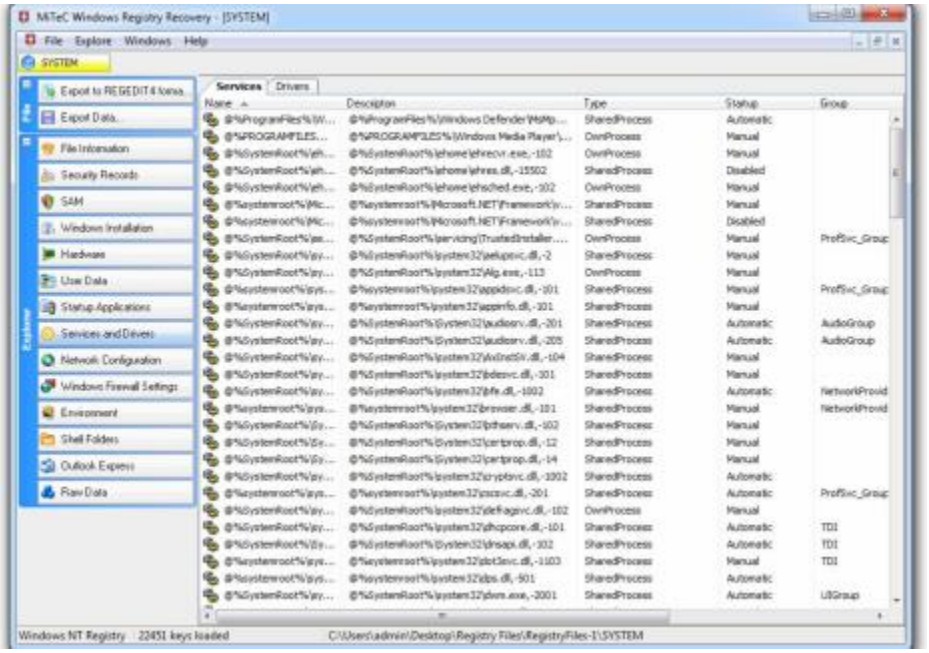
11. Navigate to the following key to identify the installation information for the versions of Windows: HKEY_LOCAL_MACHINE\TEST\Microsoft\Windows NT\CurrentVersion (product name)
12. What is the name of the Windows product? Eddie
13. What is the product ID number? 00371-868-0000007-85715
14. In what directory on the system is the operating system running (system root)? C:\Windows

15. what is the OS edition?
16. Collapse the Registry keys.
17. Click on the key named TEST. From the main menu select “File” and then choose “Unload Hive…”

1) **Activity 6: Reading Offline Registry Files with Windows Registry Recovery**

Tools: Product: MiTeC Windows Registry Recovery
WRR is part of the file pack

Instructions:
1. We’ll be working with the same files from “RegistryFiles-1.zip”
2. Download Windows Registry Recovery and extract the executable from the compressed file and place it on your desktop.
3. Launch Windows Registry Recovery.
4. Select " File " From the main menu and then “Open.” Choose the SYSTEM Registry hive.
5. On the menu on the left side of the screen click the “Services and Drivers” button. A list of services located in the SYSTEM hive will be displayed as shown below:



6. On the menu on the left side of the screen click the “Network Configuration” button.
7. In the right frame, click the “TCP/IP” tab. The network information from the Registry including the IP address will be displayed.
8. What is the IP address of the computer?

A) is it a static address or DHCP?

DhcpIPAddress	192.168.1.4

B) what is the default gateway?

DhcpDefaultGateway	192.168.1.1

9. From the main menu select “File” and then select “Open.” Browse to the SAM file.
10. After the SAM file loads, click the “SAM” button on the left side of the window.
11. On the right side of window click the “Groups and Users” tab.

12. What are the account names on this computer? MR-E
13. When was the Jimmy account last logon?

We'll continue with: MiTeC Windows Registry Recovery

1. Using the same registry in the previous activity, open NTuser.dat
2. Select the NTUSER.dat file in HKEY_USERS
3. Identify a location to save the output.
4. open to process the Registry file.
5. under raw data\software\microsoft\internet explorer\typed urls

Screen grab the url names that were last typed into this computer using internet explorer.

Value	Type	Data
url1	REG_SZ	http://www.google.com/
url2	REG_SZ	http://fedex.com/us/sports/fedexracing/employeelogin.html
url3	REG_SZ	http://www.facebook.com/
url4	REG_SZ	http://www.cnn.com/
url5	REG_SZ	http://www.apple.com/
url6	REG_SZ	https://gmail.google.com/
url7	REG_SZ	http://go.microsoft.com/fwlink/?LinkId=69157

6. Was a USB drive ever used on this computer? No

System\raw data\controlset1\usbstor\

7. From "RegistryFiles-2". We will work with the artifacts from the two files SYSTEM and NTUSER.dat.

a. What was the local area IP address assigned to the computer?

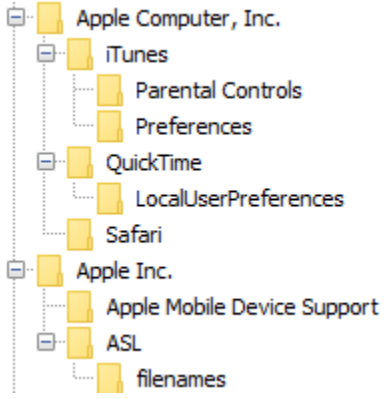
IPAddress	172.16.42.42

b. Were there at least 10 USB drives plugged into this machine

No only 4

C. What is the serial number of the iPod that was plugged into the computer?

I couldn't find the serial number



d. Search through the NTUSER.dat file. Was an Amazon Kindle connected to this computer? if so, when was the last synch? (check user settings in Raw data\software)

InstallDir	REG_SZ	C:\Users\Katie\AppData\Local\Amazon\Kindle\application
Version	REG_SZ	36179
LastUpdatedDate	REG_SZ	March 05, 2012 09:01 AM